

Data type and Conditions

1. Grocery Store Billing System

You have recently started working at a small grocery store. At the end of the day, the store manager asks you to prepare a simple report showing the names of the products sold along with their prices. This report will help the manager quickly review the sales before closing the shop.

Task:

- Use one list to store product names.
 - Use another list to store their corresponding prices.
 - Print the report in the following format:
 - [Product Name] costs ₦[Price]
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2. Student Marks Report

A teacher has collected the marks of several students after a class test. Before uploading the results, the teacher wants a simple summary showing each student's name along with their marks.

Task:

- Use one list to store student names.
 - Use another list to store their marks.
 - Print the report in the following format:
 - [Student Name] scored [Marks] marks.
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3. Restaurant Menu Display

A newly opened restaurant wants to display today's special menu along with the prices of each dish. The owner has asked you to prepare a simple program that prints the menu in an organized format.

Task:

- Use one list for food names.
 - Use another list for food prices.
 - Print the menu in the following format:
 - [Food Name] - ₹[Price]
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4. Employee Salary Report

A company is preparing a monthly salary report for its employees. The HR department wants to display each employee's name together with their salary before processing the payments.

Task:

- Use one list for employee names.
 - Use another list for salaries.
 - Print:
 - [Employee Name] will receive ₹[Salary]
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5. Favorite Movies Collection

You are creating a personal collection of your favorite movies. Since your favorite movies rarely change, you decide to store them in a data type that should not be modified accidentally.

Task:

- Store at least five movie names in a tuple.
 - Print each movie.
 - Print the total number of movies.
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6. Cricket Team Members

A school cricket coach wants to verify whether a particular student is included in the final playing squad before the tournament begins.

Task:

- Store player names in a tuple.
 - Check whether "**Rahim**" is in the team.
 - Print an appropriate message.
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7. Online Book Store

An online bookstore wants to keep track of the prices of its books. The owner asks you to create a simple program that stores book names and their prices for easy reference.

Task:

- Create a dictionary where the key is the book name and the value is its price.
 - Print:
 - [Book Name] costs ₹[Price]
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8. Student Information System

A school wants to keep basic information about a student, including their name, age, and class. The information should be stored in a single data structure for easy access.

Task:

- Create a dictionary containing Name, Age, and Class.
 - Print all the information in a readable format.
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9. Employee Contact Directory

A company wants to maintain a directory containing employee names and their phone numbers. This will help the HR department quickly find contact information when needed.

Task:

- Create a dictionary of employee names and phone numbers.
 - Print:
 - [Employee Name] : [Phone Number]
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10. Fruit Basket Organizer

A fruit shop receives fruits from different suppliers. Sometimes the same fruit arrives multiple times. The shop owner wants to keep only the unique fruit names.

Task:

- Store several fruit names in a set.
 - Print all unique fruits.
 - Print the total number of unique fruits.
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11. Classroom Attendance

A teacher records the names of students who attended today's class. The teacher wants to check whether a particular student was present.

Task:

- Store student names in a set.
 - Check whether **"Nafis"** attended the class.
 - Print an appropriate message.
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12. Welcome Message

A website greets every user after they successfully create an account. The welcome message should include the user's name to make the experience more personal.

Task:

- Store your name in a string.
 - Print:
Welcome, [Name]!
 - Print the length of your name.
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13. Username Analyzer

A social media platform wants to analyze usernames entered by new users. The system should display some basic information about the username.

Task:

- Store a username.
 - Print the first character.
 - Print the last character.
 - Print the first four characters.
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14. Weather Monitoring System

A weather station records the daily temperature and displays a simple weather message based on the recorded value.

Task:

- Input today's temperature.
- If the temperature is greater than 30, print:
It's a hot day.
- Otherwise, print:

- The weather is pleasant.
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15. Exam Result Checker

After completing the final examination, students are eager to know whether they have passed or failed. The school considers 40 as the passing mark.

Task:

- Input a student's marks.
 - If the marks are 40 or above, print:
 - Pass

 - Otherwise, print:
 - Fail
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16. Movie Ticket Eligibility

A cinema only allows customers who are at least 18 years old to watch an adult-rated movie. Before selling a ticket, the staff needs to verify the customer's age.

Task:

- Input the customer's age.
- If the age is 18 or above, print:
- Ticket Approved

- Otherwise, print:
- Not Eligible

17. Smart Door Lock

A smart home uses a digital door lock that unlocks only when the correct password is entered. The system stores the login status as either True or False.

Task:

- Store a boolean value representing whether the password is correct.
 - If the value is `True`, print:
 - Door Unlocked
 - Otherwise, print:
 - Access Denied
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18. Electricity Bill Discount

An electricity company offers a discount to customers whose monthly bill exceeds ₹5000. Your task is to determine whether a customer is eligible for the discount.

Task:

- Input the total bill amount.
 - If the bill is greater than ₹5000, print:
 - Discount Applied
 - Otherwise, print:
 - No Discount
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19. Restaurant Food Availability

A restaurant stores its menu and prices in a digital system. Whenever a customer asks for a particular food item, the system checks whether the item is available before displaying its price.

Task:

- Create a dictionary of food items and prices.
 - Input a food name.
 - If the food exists in the dictionary, print its price.
 - Otherwise, print:
 - Item Not Available
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20. Online Shopping Inventory

An online shopping website keeps track of the stock of its products. Before accepting an order, the system checks whether the selected product is currently available.

Task:

- Use one list for product names.
 - Use another list for available quantities.
 - Check the quantity of the first product.
 - If the quantity is greater than 0, print:
 - Available

 - Otherwise, print:
 - Out of Stock
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